

# Time Series — Solutions

Practice Exam — Week 10: Forecasting  
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## Problem 1 (5 points)

### Statement.

**(Forecasting a causal AR(1): point forecast, error variance, interval.)** Let  $\{X_t\}$  be the zero-mean Gaussian AR(1) process

$$X_t = \phi X_{t-1} + \varepsilon_t, \quad |\phi| < 1, \quad \varepsilon_t \stackrel{iid}{\sim} \mathcal{N}(0, \sigma^2),$$

and write  $X_{n+h}^n$  for the best linear predictor of  $X_{n+h}$  based on  $X_1, \dots, X_n$ , with  $h$ -step forecast error  $e_n(h) = X_{n+h} - X_{n+h}^n$ .

(a) Give the MA( $\infty$ ) weights  $\{\psi_j\}$  of the process. Using that the process is Gaussian (so the best linear predictor equals the conditional expectation), show that the point forecast is  $X_{n+h}^n = \phi^h X_n$ , and state its limit as  $h \rightarrow \infty$ . (1.5 pts)

(b) Show that the  $h$ -step forecast-error variance is

$$\text{Var}(e_n(h)) = \sigma^2 \frac{1 - \phi^{2h}}{1 - \phi^2},$$

and verify that  $\text{Var}(e_n(h)) \rightarrow \gamma_0$  (the marginal variance) as  $h \rightarrow \infty$ . (2 pts)

(c) Now fix  $\phi = 0.6$  and  $\sigma^2 = 4$ . Report  $X_{n+2}^n$  and the numerical forecast-error variances  $\text{Var}(e_n(1)), \text{Var}(e_n(2))$ , then give the 95% Gaussian prediction interval for  $X_{n+2}$  (use  $z_{0.975} = 1.96$ ). (1.5 pts)

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### Solution.

(a) For the causal AR(1) the MA( $\infty$ ) representation is obtained by inverting  $1 - \phi B$ :

$$X_t = \frac{1}{1 - \phi B} \varepsilon_t = \sum_{j=0}^{\infty} \phi^j \varepsilon_{t-j}, \quad \text{so} \quad \psi_j = \phi^j \quad (j \geq 0).$$

Because the process is Gaussian, the best linear predictor coincides with the conditional expectation  $\mathbb{E}[X_{n+h} \mid X_1, \dots, X_n]$ . Iterating the recursion  $X_{n+h} = \phi X_{n+h-1} + \varepsilon_{n+h}$  and taking conditional expectations (every future innovation  $\varepsilon_{n+1}, \dots, \varepsilon_{n+h}$  has conditional mean 0, being uncorrelated with the data),

$$\begin{aligned} X_{n+h}^n &= \mathbb{E}[\phi X_{n+h-1} + \varepsilon_{n+h} \mid X_1, \dots, X_n] = \phi \mathbb{E}[X_{n+h-1} \mid X_1, \dots, X_n] \\ &= \phi^2 \mathbb{E}[X_{n+h-2} \mid X_1, \dots, X_n] = \dots = \phi^h X_n. \end{aligned}$$

Since  $|\phi| < 1$ ,  $X_{n+h}^n = \phi^h X_n \xrightarrow{h \rightarrow \infty} 0$ , the process mean: the forecast *reverts to the mean*.

(b) By the  $\psi$ -weight theorem the forecast error is the sum of the unrealised future innovations,

$$e_n(h) = X_{n+h} - X_{n+h}^n = \sum_{j=0}^{h-1} \psi_j \varepsilon_{n+h-j} = \sum_{j=0}^{h-1} \phi^j \varepsilon_{n+h-j}.$$

The innovations are uncorrelated with common variance  $\sigma^2$ , so

$$\text{Var}(e_n(h)) = \sigma^2 \sum_{j=0}^{h-1} \psi_j^2 = \sigma^2 \sum_{j=0}^{h-1} \phi^{2j} = \sigma^2 \frac{1 - (\phi^2)^h}{1 - \phi^2} = \sigma^2 \frac{1 - \phi^{2h}}{1 - \phi^2},$$

using the finite geometric sum  $\sum_{j=0}^{h-1} r^j = (1 - r^h)/(1 - r)$  with  $r = \phi^2$ . As  $h \rightarrow \infty$ ,  $\phi^{2h} \rightarrow 0$  (since  $|\phi| < 1$ ), so

$$\text{Var}(e_n(h)) \rightarrow \frac{\sigma^2}{1 - \phi^2} = \gamma_0,$$

the marginal variance of the AR(1). The forecast-error variance increases monotonically in  $h$  and saturates at  $\gamma_0$ : forecasting far ahead is no better than quoting the unconditional distribution.

(c) With  $\phi = 0.6$  and  $\sigma^2 = 4$ :

$$X_{n+2}^n = \phi^2 X_n = 0.36 X_n.$$

The forecast-error variances are

$$\text{Var}(e_n(1)) = \sigma^2 \psi_0^2 = \sigma^2 = 4, \quad \text{Var}(e_n(2)) = \sigma^2(1 + \phi^2) = 4(1 + 0.36) = 4 \cdot 1.36 = 5.44.$$

(For reference  $\gamma_0 = 4/(1 - 0.36) = 4/0.64 = 6.25$ , and indeed  $4 < 5.44 < 6.25$ .) A 95% Gaussian prediction interval for  $X_{n+2}$  is

$$X_{n+2}^n \pm z_{0.975} \sqrt{\text{Var}(e_n(2))} = 0.36 X_n \pm 1.96 \sqrt{5.44} = 0.36 X_n \pm 1.96(2.3324) = 0.36 X_n \pm 4.571.$$

## Problem 2 (5 points)

### Statement.

*(The prediction equations, and one-step AR(2) forecasts.) Let  $\{X_t\}$  be a zero-mean second-order stationary process with autocovariance  $\{\gamma_\tau\}$ , and let*

$$X_{n+h}^n = \sum_{j=1}^n \beta_j X_j$$

*be its best linear predictor of  $X_{n+h}$  based on  $X = (X_1, \dots, X_n)^\top$ .*

(a) *Starting from the prediction mean square error  $S(\beta) = \mathbb{E}[(X_{n+h} - \beta^\top X)^2]$ , derive the prediction equations  $\Gamma_n \beta = \gamma_{[h]}$ , where  $\Gamma_n = [\gamma_{i-j}]_{i,j=1}^n$  and  $\gamma_{[h]} = (\gamma_{n+h-1}, \dots, \gamma_h)^\top$ , and show that this is equivalent to the orthogonality condition  $\mathbb{E}[(X_{n+h} - X_{n+h}^n) X_k] = 0$  for  $k = 1, \dots, n$ . (2 pts)*

(b) *Assuming  $\Gamma_n$  invertible, deduce the closed forms*

$$X_{n+h}^n = \gamma_{[h]}^\top \Gamma_n^{-1} X, \quad P_{n+h}^n = \gamma_0 - \gamma_{[h]}^\top \Gamma_n^{-1} \gamma_{[h]}. \quad (1.5 \text{ pts})$$

(c) *Let  $\{X_t\}$  be the causal AR(2) process  $X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + \varepsilon_t$ . Use the result of part (a) to show that the best one-step predictors based on  $X_1$  and on  $X_1, X_2$  are*

$$X_2^1 = \frac{\gamma_1}{\gamma_0} X_1 = \rho_1 X_1, \quad X_3^2 = \phi_1 X_2 + \phi_2 X_1.$$

(For the second, you need not invert  $\Gamma_2$ .)

(1.5 pts)

### Solution.

(a) Expand the prediction MSE as a quadratic in  $\beta$ , using  $\text{Var}(X) = \Gamma_n$  and  $\text{Cov}(X_{n+h}, X) = \gamma_{[h]}$  (the  $k$ -th entry of  $\text{Cov}(X_{n+h}, X)$  is  $\text{Cov}(X_{n+h}, X_k) = \gamma_{n+h-k}$ , which is exactly the  $k$ -th entry of  $\gamma_{[h]}$ ):

$$\begin{aligned} S(\beta) &= \mathbb{E}[(X_{n+h} - \beta^\top X)^2] = \text{Var}(X_{n+h} - \beta^\top X) \\ &= \gamma_0 - 2\beta^\top \text{Cov}(X_{n+h}, X) + \beta^\top \text{Var}(X) \beta = \gamma_0 - 2\beta^\top \gamma_{[h]} + \beta^\top \Gamma_n \beta. \end{aligned}$$

This is a quadratic in  $\beta$  bounded below by 0 ( $\Gamma_n$  is positive semi-definite), so its minimiser solves  $\nabla_\beta S = 0$ :

$$\nabla_\beta S = -2\gamma_{[h]} + 2\Gamma_n \beta = 0 \quad \implies \quad \boxed{\Gamma_n \beta = \gamma_{[h]}.}$$

*Equivalence to orthogonality.* Row  $k$  of  $\Gamma_n \beta = \gamma_{[h]}$  reads

$$\sum_{j=1}^n \beta_j \gamma_{j-k} = \gamma_{n+h-k}, \quad k = 1, \dots, n.$$

On the other hand  $\mathbb{E}[(X_{n+h} - \sum_j \beta_j X_j) X_k] = \text{Cov}(X_{n+h}, X_k) - \sum_j \beta_j \text{Cov}(X_j, X_k) = \gamma_{n+h-k} - \sum_j \beta_j \gamma_{j-k}$ . Setting this to 0 for each  $k$  gives exactly the same  $n$  equations. Hence “error orthogonal to every observation”  $\iff \Gamma_n \beta = \gamma_{[h]}$ : the best linear predictor is the orthogonal projection of  $X_{n+h}$  onto  $\text{span}\{X_1, \dots, X_n\}$ .

(b) If  $\Gamma_n$  is invertible then  $\beta = \Gamma_n^{-1} \gamma_{[h]}$ , so

$$X_{n+h}^n = \beta^\top X = (\Gamma_n^{-1} \gamma_{[h]})^\top X = \gamma_{[h]}^\top \Gamma_n^{-1} X$$

(using  $(\Gamma_n^{-1})^\top = \Gamma_n^{-1}$  since  $\Gamma_n$  is symmetric). For the MSE, substitute  $\beta^\top \Gamma_n \beta = \beta^\top \gamma_{[h]}$  into  $S(\beta)$ :

$$P_{n+h}^n = \gamma_0 - 2\beta^\top \gamma_{[h]} + \beta^\top \Gamma_n \beta = \gamma_0 - 2\beta^\top \gamma_{[h]} + \beta^\top \gamma_{[h]} = \gamma_0 - \beta^\top \gamma_{[h]} = \gamma_0 - \gamma_{[h]}^\top \Gamma_n^{-1} \gamma_{[h]}.$$

(c) *Based on  $X_1$  (so  $n = 1, h = 1$ ).* The prediction equation  $\Gamma_1 \beta = \gamma_{[1]}$  is the scalar equation  $\gamma_0 \beta_1 = \gamma_1$ , hence  $\beta_1 = \gamma_1 / \gamma_0 = \rho_1$  and

$$X_2^1 = \beta_1 X_1 = \frac{\gamma_1}{\gamma_0} X_1 = \rho_1 X_1.$$

*Based on  $X_1, X_2$  (so  $n = 2, h = 1$ ).* Rather than invert  $\Gamma_2$ , use the model:  $X_3 = \phi_1 X_2 + \phi_2 X_1 + \varepsilon_3$ , so the candidate predictor  $\phi_1 X_2 + \phi_2 X_1$  has error

$$X_3 - (\phi_1 X_2 + \phi_2 X_1) = \varepsilon_3.$$

The future white noise  $\varepsilon_3$  is uncorrelated with the past observations  $X_1, X_2$  (which depend only on  $\varepsilon_s, s \leq 2$ ), so

$$\mathbb{E}[(X_3 - \phi_1 X_2 - \phi_2 X_1) X_1] = \mathbb{E}[\varepsilon_3 X_1] = 0, \quad \mathbb{E}[(X_3 - \phi_1 X_2 - \phi_2 X_1) X_2] = \mathbb{E}[\varepsilon_3 X_2] = 0.$$

The error is orthogonal to both observations, so by the prediction-equations theorem of part (a) this candidate *is* the best linear predictor:

$$X_3^2 = \phi_1 X_2 + \phi_2 X_1.$$

(General fact: for an  $\text{AR}(p)$  with  $n \geq p$ , the one-step forecast is simply  $\sum_{i=1}^p \phi_i X_{n+1-i}$ .)

### Problem 3 (5 points)

**Statement.**

*(The causal-ARMA predictor via  $\psi$ -weights, and a worked ARMA(1,1).)*

(a) For a causal ARMA process with linear representation  $X_t = \sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j}$ , prove that the best linear predictor of  $X_{n+h}$  based on  $X_1, \dots, X_n$  is

$$X_{n+h}^n = \sum_{j=h}^{\infty} \psi_j \varepsilon_{n+h-j}, \quad \text{with} \quad P_{n+h}^n = \sigma^2 \sum_{j=0}^{h-1} \psi_j^2.$$

(Hint: any linear predictor based on  $X_1, \dots, X_n$  can be written  $\sum_{j \geq 0} c_j \varepsilon_{n-j}$ ; expand the MSE and minimise over the  $c_j$ .) (2 pts)

(b) Consider the causal, invertible ARMA(1,1) process

$$X_t = \frac{1}{2}X_{t-1} + \varepsilon_t + \frac{3}{10}\varepsilon_{t-1}, \quad \sigma^2 = 2.$$

Compute the first three  $\psi$ -weights  $\psi_0, \psi_1, \psi_2$  (from  $\Theta(B)/\Phi(B)$ ), and give the point forecasts  $X_{n+1}^n$  and  $X_{n+2}^n$  in terms of  $X_n$  and the residual  $\hat{\varepsilon}_n$ . (1.5 pts)

(c) Report the forecast-error variances  $P_{n+1}^n, P_{n+2}^n, P_{n+3}^n$  as exact numbers, and explain in one sentence the value  $\gamma_0$  they approach as  $h \rightarrow \infty$  (you may use  $\gamma_0 = \sigma^2(1 + 2\phi\theta + \theta^2)/(1 - \phi^2)$  for an ARMA(1,1)). (1.5 pts)

**Solution.**

(a) Consider any linear predictor  $X_{n+h}^n = \sum_{i=1}^n \beta_i X_i$ . Substituting the linear representation  $X_i = \sum_{j=0}^{\infty} \psi_j \varepsilon_{i-j}$ ,

$$X_{n+h}^n = \sum_{i=1}^n \beta_i \sum_{j=0}^{\infty} \psi_j \varepsilon_{i-j} = \sum_{j=0}^{\infty} c_j \varepsilon_{n-j}$$

for some coefficients  $c_j$ : any predictor built from  $X_1, \dots, X_n$  is a linear combination of the innovations up to time  $n$  (it “starts at  $\varepsilon_n$ ”). Now write  $X_{n+h} = \sum_{j=0}^{\infty} \psi_j \varepsilon_{n+h-j}$  and split off the future innovations  $\varepsilon_{n+1}, \dots, \varepsilon_{n+h}$  (indices  $j = 0, \dots, h-1$ ). Since the  $\varepsilon$ ’s are uncorrelated with common variance  $\sigma^2$ , the cross terms vanish and

$$\begin{aligned} \mathbb{E}[(X_{n+h} - X_{n+h}^n)^2] &= \mathbb{E}\left[\left(\sum_{j=0}^{\infty} \psi_j \varepsilon_{n+h-j} - \sum_{j=0}^{\infty} c_j \varepsilon_{n-j}\right)^2\right] \\ &= \sigma^2 \left( \sum_{j=0}^{h-1} \psi_j^2 + \sum_{j=h}^{\infty} (\psi_j - c_{j-h})^2 \right). \end{aligned}$$

The first sum collects the future innovations, which no past-based predictor can touch (it is *fixed*); the second sum matches  $\psi_j$  against  $c_{j-h}$  for the observable innovations. The second sum is minimised — to 0 — by choosing  $c_{j-h} = \psi_j$ , i.e.  $c_j = \psi_{j+h}$ . The predictor is then

$$X_{n+h}^n = \sum_{j=0}^{\infty} \psi_{j+h} \varepsilon_{n-j} = \sum_{j=h}^{\infty} \psi_j \varepsilon_{n+h-j},$$

the residual is  $X_{n+h} - X_{n+h}^n = \sum_{j=0}^{h-1} \psi_j \varepsilon_{n+h-j}$ , and the minimal MSE is

$$P_{n+h}^n = \sigma^2 \sum_{j=0}^{h-1} \psi_j^2.$$

(Operationally: *write the future and erase the future innovations.*)

(b) Here  $\phi = \frac{1}{2}$  and  $\theta = \frac{3}{10}$  (with  $\Phi(B) = 1 - \frac{1}{2}B$ ,  $\Theta(B) = 1 + \frac{3}{10}B$ ). The  $\psi$ -weights come from  $\Theta(B)/\Phi(B) = \sum_j \psi_j B^j$ :

$$\frac{1 + \frac{3}{10}B}{1 - \frac{1}{2}B} = \left(1 + \frac{3}{10}B\right) \sum_{k=0}^{\infty} \left(\frac{1}{2}\right)^k B^k.$$

Matching powers of  $B$ ,

$$\psi_0 = 1, \quad \psi_1 = \phi + \theta = \frac{1}{2} + \frac{3}{10} = \frac{4}{5}, \quad \psi_2 = \phi \psi_1 = \frac{1}{2} \cdot \frac{4}{5} = \frac{2}{5}$$

(in general  $\psi_j = (\phi + \theta)\phi^{j-1}$  for  $j \geq 1$ ). The point forecasts are obtained by zeroing future innovations in  $X_{n+h} = \frac{1}{2}X_{n+h-1} + \varepsilon_{n+h} + \frac{3}{10}\varepsilon_{n+h-1}$ :

$$\begin{aligned} X_{n+1}^n &= \frac{1}{2}X_n + \underbrace{\varepsilon_{n+1}}_{\rightarrow 0} + \frac{3}{10}\hat{\varepsilon}_n = \frac{1}{2}X_n + \frac{3}{10}\hat{\varepsilon}_n, \\ X_{n+2}^n &= \frac{1}{2}X_{n+1}^n + \underbrace{\varepsilon_{n+2}}_{\rightarrow 0} + \frac{3}{10}\underbrace{\varepsilon_{n+1}}_{\rightarrow 0} = \frac{1}{2}X_{n+1}^n = \frac{1}{4}X_n + \frac{3}{20}\hat{\varepsilon}_n. \end{aligned}$$

(For  $h \geq 2$  the MA term has vanished and the forecast obeys  $X_{n+h}^n = \frac{1}{2}X_{n+h-1}^n$ , decaying geometrically to the mean 0.)

(c) With  $\sigma^2 = 2$  and  $\psi_0 = 1, \psi_1 = \frac{4}{5}, \psi_2 = \frac{2}{5}$ , the forecast-error variances  $P_{n+h}^n = \sigma^2 \sum_{j=0}^{h-1} \psi_j^2$  are

$$P_{n+1}^n = \sigma^2 \psi_0^2 = 2 \cdot 1 = 2,$$

$$P_{n+2}^n = \sigma^2 (\psi_0^2 + \psi_1^2) = 2 \left( 1 + \frac{16}{25} \right) = 2 \cdot \frac{41}{25} = \frac{82}{25} = 3.28,$$

$$P_{n+3}^n = \sigma^2 (\psi_0^2 + \psi_1^2 + \psi_2^2) = 2 \left( 1 + \frac{16}{25} + \frac{4}{25} \right) = 2 \cdot \frac{45}{25} = \frac{18}{5} = 3.6.$$

These increase toward the marginal variance

$$\gamma_0 = \sigma^2 \frac{1 + 2\phi\theta + \theta^2}{1 - \phi^2} = 2 \cdot \frac{1 + 2 \cdot \frac{1}{2} \cdot \frac{3}{10} + \frac{9}{100}}{1 - \frac{1}{4}} = 2 \cdot \frac{1 + \frac{3}{10} + \frac{9}{100}}{\frac{3}{4}} = 2 \cdot \frac{139/100}{3/4} = \frac{278}{75} \approx 3.707,$$

which is exactly  $\sigma^2 \sum_{j=0}^{\infty} \psi_j^2$ : as  $h \rightarrow \infty$  the forecast loses all information and the error variance saturates at the unconditional variance  $\gamma_0$  ( $2 < 3.28 < 3.6 < 3.707$ ).

#### Problem 4 (5 points)

**Statement.**

*(Revision of Week 9: VAR stability and the diagonal VAR(1) covariance.)*

(a) Consider the bivariate VAR(2) process  $X_t = \Phi_1 X_{t-1} + \Phi_2 X_{t-2} + \varepsilon_t$  with

$$\Phi_1 = \begin{pmatrix} 0.4 & 0 \\ 0.5 & 0.9 \end{pmatrix}, \quad \Phi_2 = \begin{pmatrix} 0 & 0 \\ 0.3 & -0.2 \end{pmatrix}.$$

Form the reverse characteristic polynomial  $\det(I_2 - \Phi_1 z - \Phi_2 z^2)$ , exploit the lower-triangular structure to factor it, find all its roots, and decide whether the VAR(2) is stable. State the companion-matrix ( $F$ ) form of the stability test and explain why it gives the same conclusion. (3 pts)

(b) Now take the diagonal VAR(1) process  $X_t = \Phi X_{t-1} + \varepsilon_t$  with

$$\Phi = \text{diag}\left(\frac{1}{2}, \frac{1}{4}\right), \quad \{\varepsilon_t\} \sim \text{WN}(0, I_2).$$

Write its MA( $\infty$ ) representation and compute the stationary covariance  $\Gamma_0$  and the lag-1 matrix  $\Gamma_1$  in closed form (exact fractions). (1.5 pts)

(c) For the process of part (b), what is the coherence  $r_{1,2}(f)$  between the two channels at every frequency, and why? (0.5 pts)

**Solution.**

(a) Build  $M(z) = I_2 - \Phi_1 z - \Phi_2 z^2$ :

$$M(z) = \begin{pmatrix} 1 - 0.4z & 0 \\ -0.5z - 0.3z^2 & 1 - 0.9z + 0.2z^2 \end{pmatrix}.$$

This matrix is lower triangular, so its determinant is the product of the diagonal entries (the off-diagonal terms  $\Phi_1^{(2,1)}, \Phi_2^{(2,1)}$  do not enter):

$$\det M(z) = (1 - 0.4z)(1 - 0.9z + 0.2z^2).$$

*Roots.* The first factor gives  $z = 1/0.4 = 2.5$ . The quadratic  $0.2z^2 - 0.9z + 1 = 0$  has

$$z = \frac{0.9 \pm \sqrt{0.81 - 0.8}}{0.4} = \frac{0.9 \pm 0.1}{0.4} \implies z = 2.5 \text{ or } z = 2.$$

So the three roots are  $z \in \{2.5, 2.5, 2\}$ , all with  $|z| > 1$  (strictly outside the unit circle). Hence the VAR(2) is **stable** (and therefore stationary). *Sanity check:*  $\det M(0) = \det I_2 = 1$ , as required.

*Companion form.* Stacking  $Y_t = (X_t^\top, X_{t-1}^\top)^\top \in \mathbb{R}^4$  turns the VAR(2) into the VAR(1)  $Y_t = FY_{t-1} + U_t$  with

$$F = \begin{pmatrix} \Phi_1 & \Phi_2 \\ I_2 & 0 \end{pmatrix} = \begin{pmatrix} 0.4 & 0 & 0 & 0 \\ 0.5 & 0.9 & 0.3 & -0.2 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}.$$

The companion test is “all eigenvalues of  $F$  strictly inside the unit circle”. It is the reciprocal of the root test, because  $\det(I_2 - \Phi_1 z - \Phi_2 z^2) = z^4 \det(z^{-1}I_4 - F)$  up to a constant, so the eigenvalues of  $F$  are the reciprocals  $1/z_i$  of the roots. Here  $\lambda = 1/2.5 = 0.4$ ,  $1/2.5 = 0.4$ ,  $1/2 = 0.5$  (and a structural  $\lambda = 0$  from the rank-deficient  $\Phi_2$ ), all with  $|\lambda| < 1$ : *same* conclusion, stable.

(b) Since  $\Phi = \text{diag}(\frac{1}{2}, \frac{1}{4})$  has spectral radius  $\frac{1}{2} < 1$ , the VAR(1) is stable and back-substitution gives the matrix geometric (MA( $\infty$ )) filter

$$X_t = \sum_{j=0}^{\infty} \Phi^j \varepsilon_{t-j}, \quad \Phi^j = \text{diag}(2^{-j}, 4^{-j}).$$

By the VAR(1) covariance formula  $\Gamma_0 = \sum_{k \geq 0} \Phi^k \Sigma (\Phi^k)^\top$  with  $\Sigma = I_2$  (everything diagonal, so the channels decouple into independent AR(1)'s),

$$\Gamma_0 = \sum_{k=0}^{\infty} \text{diag}(4^{-k}, 16^{-k}) = \text{diag}\left(\frac{1}{1 - \frac{1}{4}}, \frac{1}{1 - \frac{1}{16}}\right) = \boxed{\text{diag}\left(\frac{4}{3}, \frac{16}{15}\right)}.$$

(Equivalently, the Lyapunov equation  $\Gamma_0 = \Phi \Gamma_0 \Phi^\top + I_2$  gives componentwise  $g_{jj} = \phi_j^2 g_{jj} + 1 \Rightarrow g_{jj} = 1/(1 - \phi_j^2)$ , i.e.  $1/(1 - \frac{1}{4}) = \frac{4}{3}$  and  $1/(1 - \frac{1}{16}) = \frac{16}{15}$ .) The lag-1 matrix follows from Yule–Walker  $\Gamma_1 = \Phi \Gamma_0$ :

$$\Gamma_1 = \text{diag}(\frac{1}{2}, \frac{1}{4}) \text{diag}(\frac{4}{3}, \frac{16}{15}) = \boxed{\text{diag}(\frac{2}{3}, \frac{4}{15})}.$$

(c) Both  $\Phi$  and  $\Sigma = I_2$  are diagonal, so by the formula in (b) *every*  $\Gamma_\tau$  is diagonal: all off-diagonal cross-covariances vanish at every lag. Hence the cross-spectrum  $S_{1,2}(f) = \sum_\tau \gamma_\tau^{(1,2)} e^{-2\pi i f \tau} \equiv 0$ , and the coherence

$$r_{1,2}(f) = \frac{S_{1,2}(f)}{\sqrt{S_{1,1}(f)S_{2,2}(f)}} = 0 \quad \text{for all } f.$$

The two channels are completely uncoupled: the off-diagonal entries of  $\Phi$  (here zero) are exactly the cross-channel dynamics, and there is no shared innovation ( $\Sigma$  diagonal), so nothing links the series at any frequency.